MINI PROJECT

For this mini-project, we propose 3 project variants and you will only do the one assigned to your group. Each variant has a practical part that will require you to write code and provide results in the form of images and plots, and a theoretical part where you will need to elaborate on concepts based on your knowledge and report your reasoning. You will need to address both parts and include them in your report and video recording.

Make sure that you properly familiarize yourself with the instructions before you proceed.

The deliverables should include **a written report, a code repository**, **and a recording** which should last **between 4 and 5 minutes (sharp).**

Recordings Guidelines:

- Use Zoom recording option
- Presenting flow:
    - Briefly present the dataset and the motivation of the mini project
    - Answer succinctly questions in the given order first starting from **part 1**'s practicals then theoreticals followed by **part 2** in a similar fashion
- Follow all the other indications given in the "Miniproject guideline" document that has been previously shared with you.

Written reports Guidelines:

- **5 pages max** (including images) + 1 extra page for the appendix, where you can add any additional figure you might want to use to support your results. The document should be single column, font-size 11px in Arial.
    - We suggest that you address each part separately; for each part, discuss and report the results of the practical, then provide answers to the theoretical questions.
    - Include a brief list of the contributions of each group member.
- Please, **do not** upload datasets, images beyond the ones in the report, or any additional content.
- **Submit the code** you used for solving the mini-project with the structure and naming described in the "Miniproject guideline" document that has been previously shared with you.

All deliverables should be submitted on Moodle.

Deliverables are due on **Thursday 5th November at 16:00**.
Note that this is a sharp deadline, no extensions are allowed. We therefore highly recommend you to take into account all possible uploading/formatting issues

that may incur delays in submission by submitting early. Repeated submission is allowed, we will only consider the last submission from each group.

Task description

As a researcher you are interested in studying the brain neural activity related to *language* tasks.

The dataset that you will use is a subset taken from the Human Connectome Project, a publicly available neuroimaging dataset. It is accessible through this link, and is acquired during a language task experiment of a single subject.

In particular, this task was adapted from a previously developed one by Binder and colleagues (Binder et al. 2011 [1]) with the goal to investigate and map language areas**.** We report here relevant information on the task, from the HCP website *"The task consists of two runs that each interleave 4 blocks of a story task and 4 blocks of a math task. The lengths of the blocks vary (average of approximately 30 seconds), but the task was designed so that the math task blocks match the length of the story task blocks, with some additional math trials at the end of the task to complete the 3.8 minute run as needed. The story blocks present participants with brief auditory stories (5-9 sentences) adapted from Aesop's fables, followed by a 2-alternative forced-choice question that asks participants about the topic of the story. The example provided in the original Binder paper (p. 1466) is "For example, after a story about an eagle that saves a man who had done him a favor, participants were asked, "Was that about revenge or reciprocity?" The math task also presents trials auditorially and requires subjects to complete addition and subtraction problems. The trials present subjects with a series of arithmetic operations (e.g., "fourteen plus twelve"), followed by "equals" and then two choices (e.g., "twenty-nine or twenty-six"). Participants push a button to select either the first or the second answer. The math task is adaptive to try to maintain a similar level of difficulty across participants".* The timing of the events is provided in the *event_*.csv* files found in the dataset structure.

Here you find the final structure of the folder

- **T1w**
  - T1w.nii.gz
- **dataset_description.md**
- **fMRI**
  - **tfMRI_LANGUAGE_LR**
    - tfMRI_LANGUAGE_LR_task_onsets.csv
    - tfMRI_LANGUAGE_LR.nii.gz

Part 1 (Suggested time allocation: 13th October - 29th October)

Practicals - Part 1

Using what you learnt in the lab sessions

1) **Structural preprocessing**: Apply the following steps using the methods of your choice (e.g., *fMRIPrep, FSL*-based, python-based):
    a) **Skull-stripping**. Report and plot the original T1w image and the skull-stripped brain to visually assess the quality of the brain extraction.
    b) **Segmentation**. Report and plot the segmented tissues on the brain.
    c) **Registration.** Register your results bringing them to MNI space. Report a plot of the T1 alignment with MNI before and after this step.

2) **Functional preprocessing**: Apply the following steps using the methods of your choice (e.g., *fMRIPrep, FSL*-based, python-based):
    a) **Motion correction** (you can take care of the irregular volumes the way you want). Report and plot the framewise displacement and comment on the results.
    b) **Gaussian smoothing** with an appropriate FWHM. Report the brain plot of a single fMRI volume of your choice before and after smoothing.
    c) **Validate the alignment between anatomical and functional data**. Overlay the fMRI to the structural data that you previously registered to MNI space. Report a plot of the overlay at one fMRI volume and section of your choice and comment on the general quality of the alignment. Is there a fMRI preprocessing step that can be avoided given what you see? Justify your answer based on the plot.

Briefly justify in your report the purpose of each of the structural and functional preprocessing steps.

3) **GLM:**
    The goal of the GLM is to investigate the activation of the brain in the *math* and *story* tasks.
    a) Build an appropriate experimental design matrix that you will use in the following GLM analysis.
    b) Compute and plot the correlation matrix between the regressors of the design matrix and comment on the results.
    c) Run the GLM at the level you deem reasonable and report the statistical maps of each of the task-related regressors. Briefly comment on the statistical maps.
    d) Using the results of the GLM analysis and a contrast vector of your choice, create the activation maps corresponding to the story vs math

contrast. Please include in the report the contrast vector used and the contrast map you obtained.

e) Discuss which brain regions and networks are more activated during numerical than semantic processing, and which are more activated in the opposite contrast. As brain activity at rest can be separated into known functional networks [2], comment the results in terms of functional networks. Report known functional networks found within the statistical results and brain activation maps. You can either observe their overlaps or use brain atlas to inform your answer.

f) **[Hard]** Based on the literature, some regions should stand out as particularly relevant to *semantic* language processing. Are these regions present in your contrast maps? If not, discuss at least two possible reasons that could explain their absence.

Theoreticals - Part 1

Answer the following questions, and justify your answers (you do not need to write nor provide any code for this, only present your reasoning):

1) Discuss how you would design a contrast to investigate the following tasks, and how you would interpret the resulting maps. You want to investigate
    a) the brain areas stimulated by either story or math tasks,
    b) the differences in brain response between a question and an answer in either task,
    c) which areas are more involved in task processing than in task preparation.
2) Which statistical test would you use to assess the importance of having the *story* regressor in your GLM model?
3) Assume that you want to increase the statistical power of the GLM analysis you performed above by adding a second run with the same experimental blocks and regressor types. Explicitly state the hypothesis you would test and describe how you would use the two runs to perform the analysis.

Part 2 (Suggested time allocation: 29th October - 5th November)

Note that this part is split into 3 variants, you are assigned to one of the following variants according to your group letter.

Now you proceed to do a different analysis by employing multivariate pattern analysis methods (MVPA). The goal is to find spatial maps in a data-driven manner (no GLM) and to investigate which maps would include the regions found in the GLM analysis.

You know that different methods exist to find these patterns, and you will be required to use one of them based on the project variant your group is assigned to:

- Principal Component Analysis (**variant 1**)
- Independent Component Analysis (**variant 2**)
- K-means clustering (**variant 3**)

Practicals - Part 2

Variant 1

1. Apply PCA on the subject fMRI runs, considering one **timepoint** as one sample.
2. Select a number of components of your choice to keep while providing a justification for your choice. Show each of the selected PCA loadings separately, in axial view only.
3. Compute the correlation between the (temporal) scores associated to the first 10 components and the regressors of the GLM you performed in Part 1 (including all regressors that you deem appropriate). Report the correlation matrix and comment on the results, relating them to the spatial maps obtained at point 2.

Variant 2

1. Apply ICA on the subject fMRI runs, considering one **timepoint** as one sample.
2. Select a number of independent components to retain and provide a justification for your choice. Display the spatial map of each selected component separately, in axial view only.
3. Compute the correlation between the timecourses associated to the first 10 independent components (in the order returned by the ICA algorithm) and the regressors of the GLM performed in Part 1 (including all regressors that you deem appropriate). Report the correlation matrix and comment on the results, relating them to the spatial maps obtained at point 2.

Variant 3

1. Apply temporal K-means clustering on the subject fMRI runs, considering one **timepoint** as one sample.
2. Select an appropriate number of clusters and justify your choice. Display the spatial pattern represented by each cluster centroid separately, overlaid on the anatomical image in axial view only.
3. For the first 10 clusters, construct a binary time course indicating the timepoints assigned to that cluster and compute its correlation with the GLM regressors (including all regressors that you deem it appropriate). Report the resulting correlation matrix and comment on the results, relating them to the spatial maps obtained at point 2.

Theoreticals - part 2

Answer the following questions, and justify your answers (you do not need to write nor provide any code for this, only report your reasoning):

1) How would your components change in terms of dimensionality and interpretation if you were to use one voxel as one sample? What would be their interpretation
2) Is it conceptually appropriate to compare the results from the GLM obtained in Part 1 to the temporal patterns obtained in your multivariate pattern analysis method in Part 2? What are the common points and the differences between the two approaches? When would you apply one and when would you apply the other? Please, justify your answers.

References

[1] Binder, J. R., Gross, W. L., Allendorfer, J. B., Bonilha, L., Chapin, J., Edwards, J. C., ... & Weaver, K. E. (2011). Mapping anterior temporal lobe language areas with fMRI: a multicenter normative study. *Neuroimage*, *54*(2), 1465-1475.

[2] Yeo, BT Thomas, Fenna M. Krienen, Jorge Sepulcre, Mert R. Sabuncu, Danial Lashkari, Marisa Hollinshead, Joshua L. Roffman et al. "The organization of the human cerebral cortex estimated by intrinsic functional connectivity." *Journal of neurophysiology* (2011).